Session 6: Loops (For, While, Do-While)Topics:Difference between entry-controlled and exit-controlled loops.Use cases (e.g., menu-driven programs).Example: Countdown timer using loops.

1) **Write a program that prints the numbers from 10 down to 1 using a for loop, similar to a countdown timer.**

```
#include<stdio.h>

int main(){

   int i;
   printf("countdwon timer:\n");

   for(i=10; i>=1; i--){
      printf("%d\n",i);



   }
   return 0;

}
```

Output -

**2) Create a menu-driven console app that lets the user: 1) View your favorite 3 IPL teams, 2) Add a new team, 3) Exit. Use a while loop to keep showing the menu until the user chooses Exit.

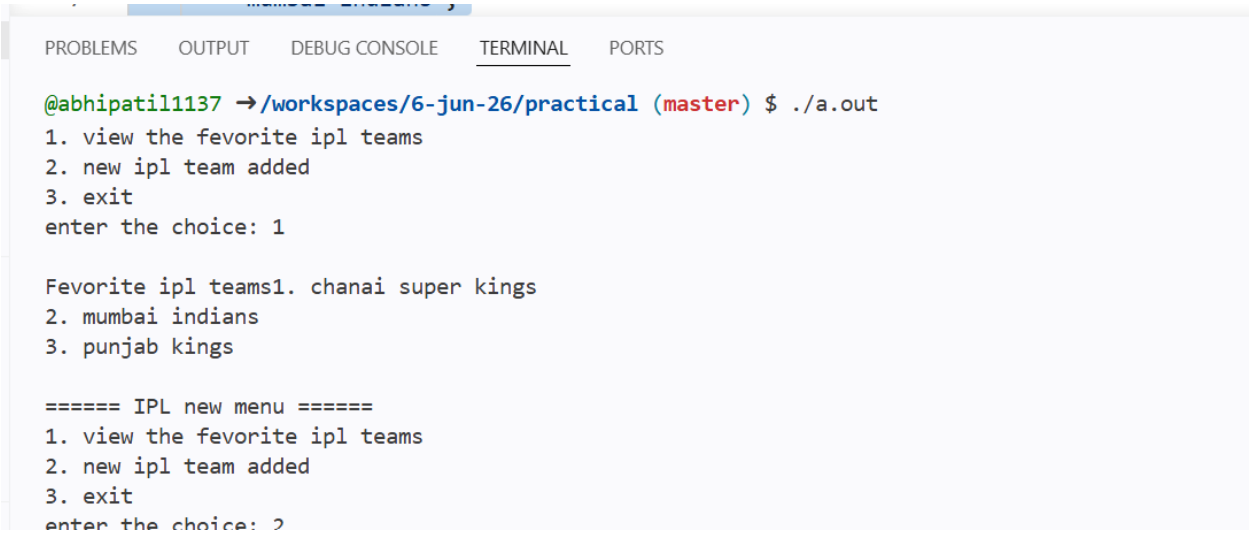
Hint: Use input() (or Scanner in Java) to get the user's choice each time.**

```
#include<stdio.h>

int main(){

   char teams[4][30]={
      "chanai super kings",
      "mumbai indians",
      "punjab kings"
   };
   int choice;
   int added=0;

   while (1)
   {
    printf("\n====== IPL new menu ======\n");
    printf("1. view the fevorite ipl teams\n");
    printf("2. new ipl team added\n");
    printf("3. exit\n");
```

```
 printf("enter the choice: ");
 scanf("%d",&choice);

 switch (choice)
 {
 case 1:
   printf("\nFevorite ipl teams");
   printf("1. %s\n",teams[0]);
   printf("2. %s\n",teams[1]);
   printf("3. %s\n",teams[2]);

   if(added)
   {
      printf("4. %s\n",teams[3]);
   }

   break;
   case 2:
   if(!added){
      printf("enter the new ipl team\n");
      scanf("%[^\n]",teams[3]);
      added=1;
      printf("team added succsefull\n");
   } else{
      printf("only one team can be added\n");

   }
break;
   case 3:
   printf("exite the progaram \n");
   return 0;

 default:
 printf("invalid choice  plsease try again\n");


 }
}
```

```
    return 0;
}
```

Output-



**3) Build a 'Guess the Song' game like Spotify — the program randomly picks a song name from a list and asks the user to guess it. Use a do-while loop so the user can keep guessing until they get it right.

Constraint: Use at least 3 song names of your choice.**

```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
#include <time.h>

int main() {
    char songs[3][30] = {
        "Shape of You",
        "Believer",
        "Perfect"
    };

    char guess[30];
    int random;
```

```
    // Generate random song
    srand(time(0));
    random = rand() % 3;

    printf("===== Guess the Song Game =====\n");
    printf("Available songs are one of these:\n");
    printf("- Shape of You\n");
    printf("- Believer\n");
    printf("- Perfect\n\n");

    do {
        printf("Enter your guess: ");
        scanf(" %[^\n]", guess);

        if (strcmp(guess, songs[random]) == 0) {
            printf("🎉 Correct! You guessed the song.\n");
        } else {
            printf("❌ Wrong guess! Try again.\n");
        }

    } while (strcmp(guess, songs[random]) != 0);

    return 0;
}
```

Output-

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

@abhipatil1137 ➜ /workspaces/6-jun-26/practical (master) $ ./a.out

Enter your guess: perfect
❌ Wrong guess! Try again.
Enter your guess: believr
❌ Wrong guess! Try again.
Enter your guess: shape of you
❌ Wrong guess! Try again.
Enter your guess: guess
❌ Wrong guess! Try again.
Enter your guess: believer
❌ Wrong guess! Try again.
Enter your guess: Believer
🎉 Correct! You guessed the song.
```

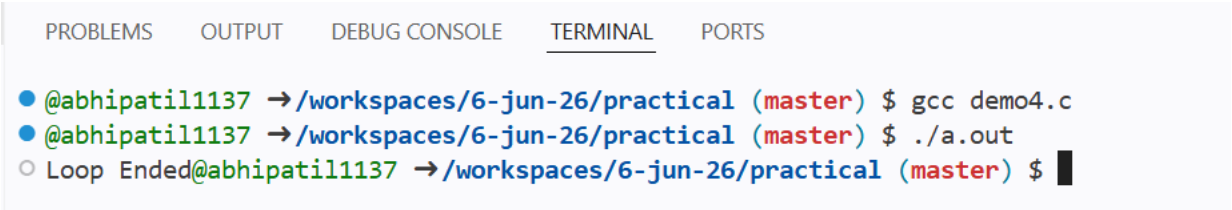
4) Explain with your own example the difference between entry-controlled and exit-controlled loops by writing a short code snippet for each (for/while vs do-while) and describing what happens if the loop condition is false at the start.

Entry-Controlled Loop (`for`, `while`)	Exit-Controlled Loop (`do-while`)
The condition is checked **before** the loop body executes.	The condition is checked **after** the loop body executes.
If the condition is false at the beginning, the loop body **does not execute**.	The loop body **executes at least once**, even if the condition is false initially.

Code1-Entry-Controlled Loop (while)

```
#include <stdio.h>

int main() {
   int num = 5;

   while (num < 5) {
      printf("%d\n", num);
      num++;
   }

   printf("Loop Ended");
   return 0;
}
```

Output –



Code-Exit-Controlled Loop (do-while)

```
#include <stdio.h>

int main() {
   int num = 5;

   do {
      printf("%d\n", num);
      num++;
   } while (num < 5);

   printf("Loop Ended");
   return 0;
}
```

Output-